

Directional Derivative and the Gradient Vector – Part II

Trim Ch 11, S. 12.8, 12.4

Outline:

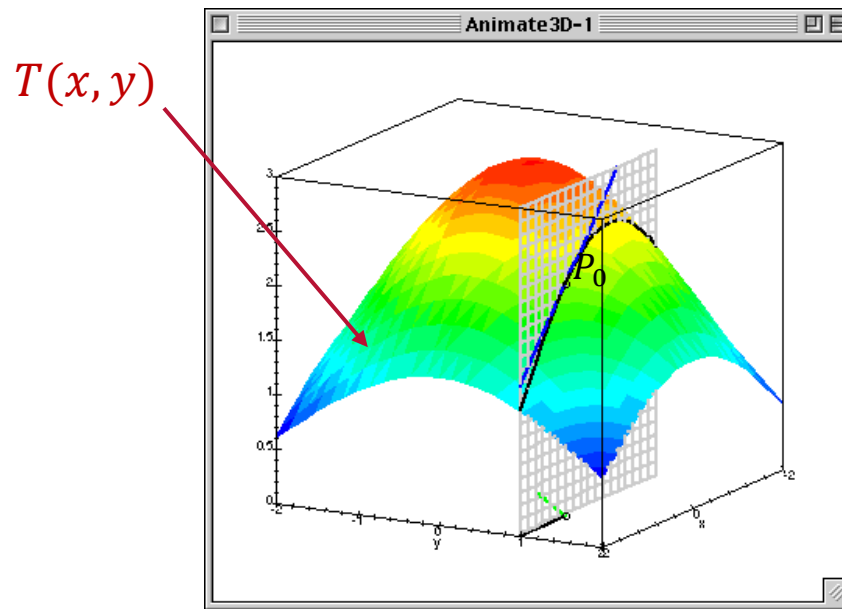
1. Properties of the directional derivative and the gradient vector



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Directional Derivative – Physical Interpretation

If $T(x, y)$ is the temperature at each point $P_0(x_0, y_0)$ of the surface, defined over a region R ,



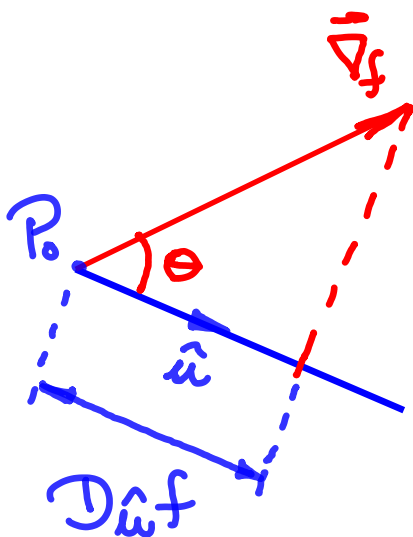
The directional derivative is the **instantaneous rate of change** of temperature at P_0 **stepping off in the direction of $\hat{u}$**

$$D_{\hat{u}}T|_{P_0} = \left. \frac{dT}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}T|_{P_0} \cdot \hat{u}$$

Directional Derivative in the Direction of $\hat{u}$

The directional derivative of a function $f(x, y)$ at P_0 in the direction of $\hat{u}$ gives us insight into some of the properties of the gradient vector,

$$D_{\hat{u}}f|_{P_0} = \left. \frac{df}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}f|_{P_0} \cdot \hat{u}$$



$$|D_{\hat{u}}f| = |\vec{\nabla}f| |\hat{u}| \cos\theta$$

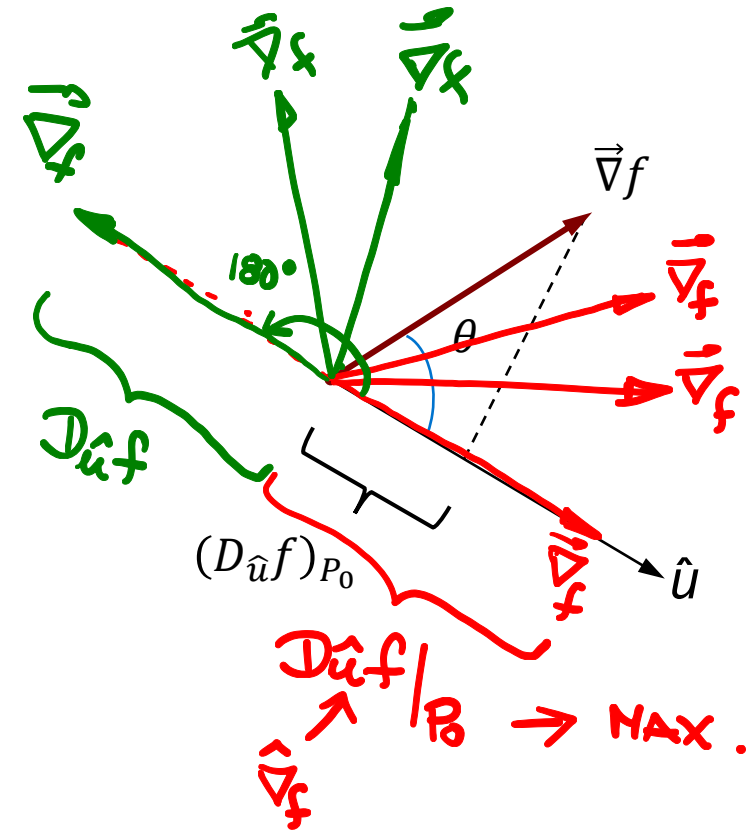
$\hat{u} = 1$

$$D_{\hat{u}}f = |\vec{\nabla}f| \cos\theta$$

The directional derivative $D_{\hat{u}}f|_{P_0}$ is the component of the gradient vector at P_0 in the direction of $\hat{u}$.



Maximum and Minimum Rate of Change



IF $\theta = 0$, $\cos \theta = 1$

$D_{\hat{u}} f$ IS MAXIMUM

$$\therefore D_{\hat{u}} f = |\vec{\nabla} f|$$

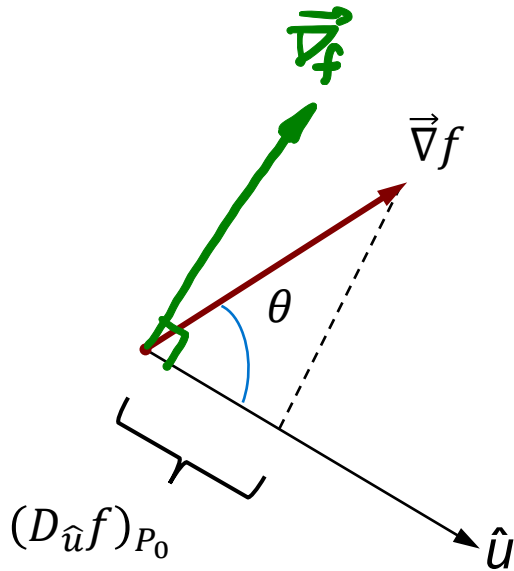
IF $\theta = 180^\circ$, $\cos \theta = -1$

$D_{\hat{u}} f$ IS ALSO MAXIMUM

$$D_{\hat{u}} f = -|\vec{\nabla} f|$$

f INCREASES (DECREASES) MOST RAPIDLY IN THE DIRECTION OF $\vec{\nabla} f$ ($-\vec{\nabla} f$)

Direction of Zero Change



$$\text{IF } \theta = 90^\circ, \cos \theta = 0$$

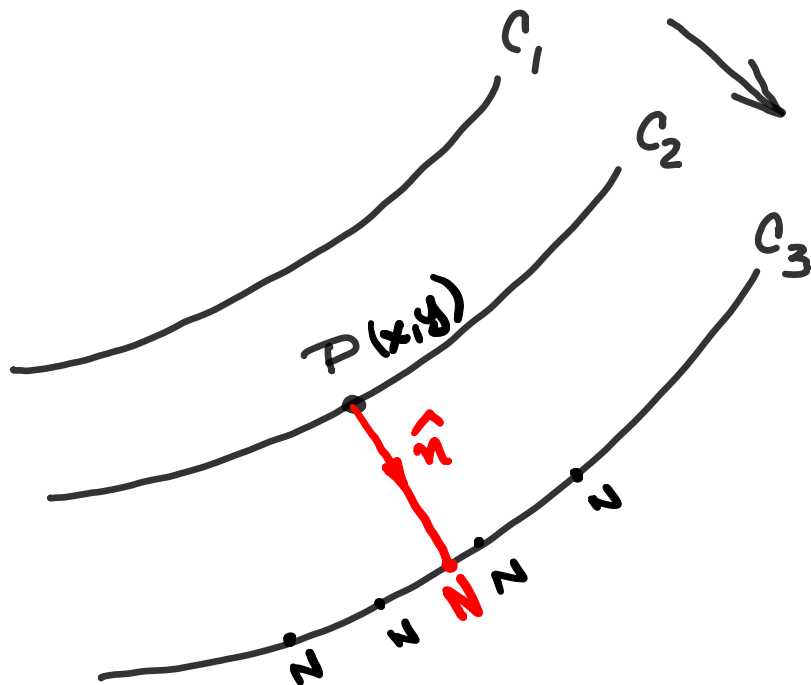
$$\therefore D_{\hat{u}}f = 0$$

ANY DIRECTION $\hat{u} \perp \vec{\nabla} f$ IS
A DIRECTION OF ZERO
CHANGE!



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

$\vec{\nabla} f$ can be observed on a contour map:



1. When is the rate of change from P to N the largest?

WHEN PN IS THE SMALLEST

2. When is the distance between PN the smallest?

WHEN "N" LIES ON THE NORMAL TO THE CONTOUR AT "P"

3. If f increases ~~in~~ most rapidly in the direction of $\vec{\nabla} f$, at P, in which direction will $\vec{\nabla} f$ be pointing?

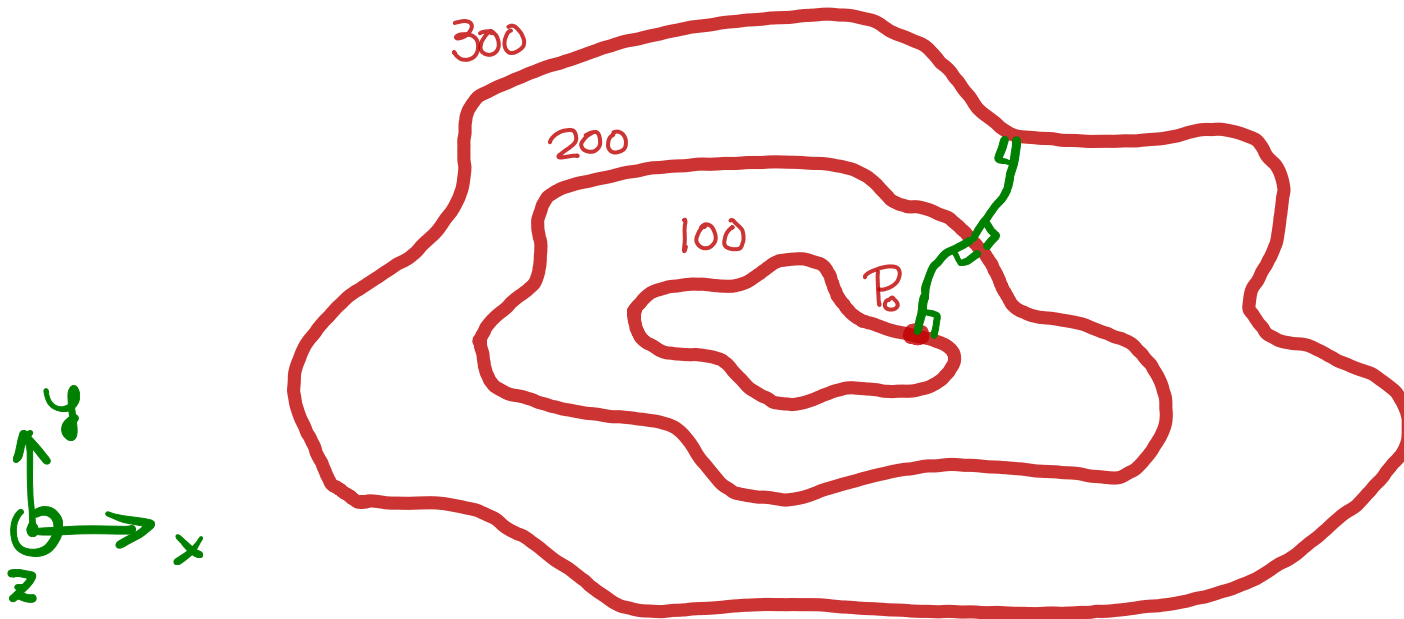
⊥ TO THE CURVE AT P

$\vec{\nabla} f$ is in the direction of the outward normal $\hat{n}$ to the level curve.



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

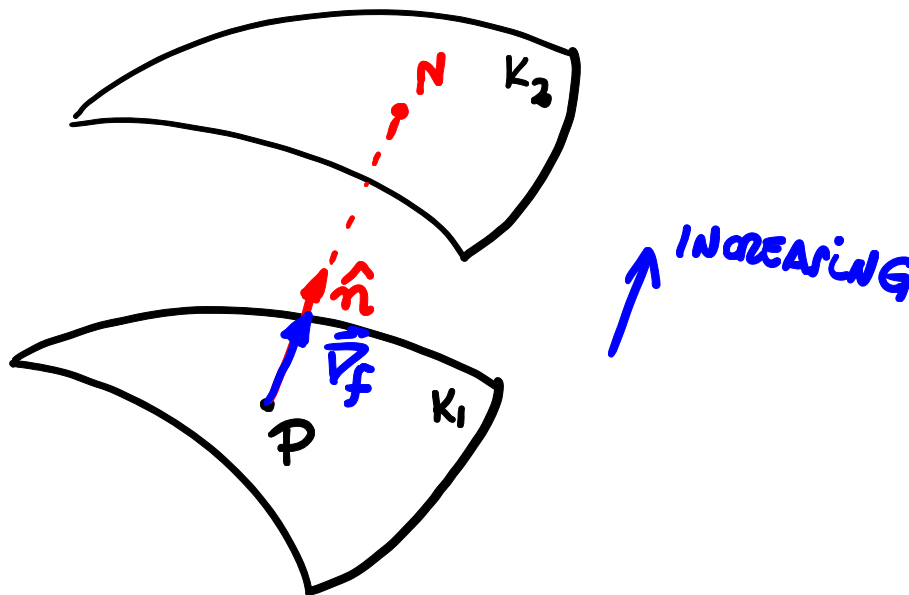
How do I travel from an elevation of 100 meters to an elevation of 300 meters the fastest?



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

$\vec{\nabla} f$ can also be observed on a family of surface levels:

$$f(x, y, z) = k$$



The gradient represents, in magnitude and direction, the greatest rate of increase (decrease) of f with distance



In our example ...

The gradient of the function $f(x, y) = x^3 - 3xy + 4y^2$ was found as,

$$\vec{\nabla} f = (3x^2 - 3y)\hat{i} + (-3x + 8y)\hat{j} \quad \leftarrow \textcircled{I}$$

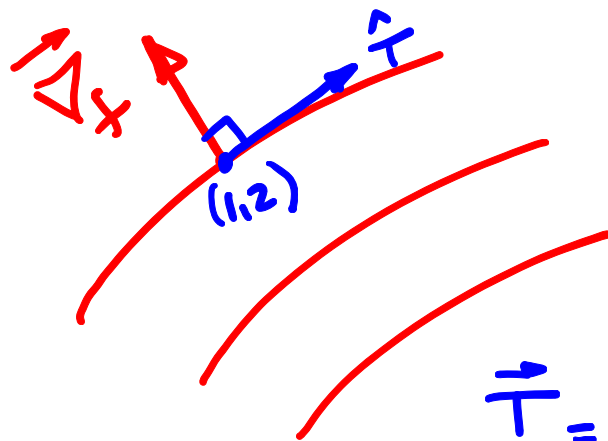
a. What is the direction of maximum increase at the point (1,2)

IN $\textcircled{I}$
FOR (1,2)

$$\vec{\nabla} f \big|_{(1,2)} = (-3, 13)$$

$$\hat{\nabla} f = \frac{(-3, 13)}{\sqrt{3^2 + 13^2}}$$

b. What is the direction of zero change at (1,2)



DIRECTION OF $\hat{T}$ (ZERO CHANGE)

$$(\vec{\nabla} f \cdot \hat{T}) \big|_{(1,2)} = 0$$

$$\vec{T} = (13, 3) \rightarrow \hat{T} = \frac{(13, 3)}{\sqrt{174}}$$



Problem

Calculate the directional derivative of the function,

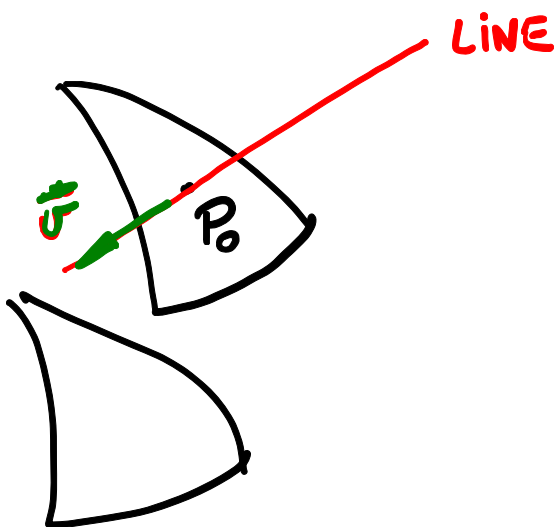
$$f(x, y, z) = x^2y + y^2z + z^2x$$

at the point $(1, -1, 0)$ along the line,

$$x + 2y + 1 = 0$$

$$x - y + 2z = 2$$

in the direction of decreasing z .



DIR. OF LINE OF INTERSECTION OF 2 PLANES

$$\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 1 & -1 & 2 \end{vmatrix} = \dots =$$

$$\vec{v} = (4, -2, -3)$$

Problem

$$D_{\hat{u}} f(1, -1, 0) = \vec{\nabla} f \Big|_{(1, -1, 0)} \cdot \underbrace{\hat{u}}_{\substack{(4, -2, -3) \\ \sqrt{16+4+9}}}$$

$$\vec{\nabla} f = (f_x, f_y, f_z)$$

$$= (2xy + z^2, x^2 + 2yz, y^2 + 2zx)$$

$$\vec{\nabla} f \Big|_{(1, -1, 0)} = (-2, 1, 1)$$

$$D_{\hat{u}} f(1, -1, 0) = (-2, 1, 1) \cdot \frac{(4, -2, -3)}{\sqrt{29}}$$

$$= \frac{-13}{\sqrt{29}}$$



Problem

What is the direction of maximum increase of $f(x, y, z)$ at $(1, -1, 0)$?

$$\vec{\nabla} f(1, -1, 0) = (-2, 1, 1)$$

What is the direction at which we need to travel for zero change?

NO CHANGE IF TRAVELING ALONG THE
LEVEL SURFACE AT P_0

→ DIRECTION OF TANGENT PLANE !

NEXT LECTURE



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Summary

- **The directional derivative is the instantaneous rate of change of a function $f(x, y)$ at P stepping off in the direction of $\hat{u}$**
- The directional derivative is the component of the gradient vector in the direction of $\hat{u}$
- The function $f(x, y)$ increases (decreases) most rapidly in the direction of $\vec{\nabla}f$ ($-\vec{\nabla}f$)
- Any direction $\hat{u}$ perpendicular to $\vec{\nabla}f$ is a direction of zero change
- The gradient represents, in magnitude and direction, the greatest rate of increase/decrease of $f(x, y)$ with distance